

The Impact of Microglia on Neurodevelopment and Brain Function in Autism.

PMID: 38255315 | Indexed in PubMed

Microglia, as one of the main types of glial cells in the central nervous system (CNS), are widely distributed throughout the brain and spinal cord. The normal number and function of microglia are very important for maintaining homeostasis in the CNS. In recent years, scientists have paid widespread attention to the role of microglia in the CNS. Autism spectrum disorder (ASD) is a highly heterogeneous neurodevelopmental disorder, and patients with ASD have severe deficits in behavior, social skills, and communication. Most previous studies on ASD have focused on neuronal pathological changes, such as increased cell proliferation, accelerated neuronal differentiation, impaired synaptic development, and reduced neuronal spontaneous and synchronous activity. Currently, more and more research has found that microglia, as immune cells, can promote neurogenesis and synaptic pruning to maintain CNS homeostasis. They can usually reduce unnecessary synaptic connections early in life. Some researchers have proposed that many pathological phenotypes of ASD may be caused by microglial abnormalities. Based on this, we summarize recent research on microglia in ASD, focusing on the function of microglia and neurodevelopmental abnormalities. We aim to clarify the essential factors influenced by microglia in ASD and explore the possibility of microglia-related pathways as potential research targets for ASD.

Migration and Phagocytic Ability of Activated Microglia During Post-natal Development is Mediated by Calcium-Dependent Purinergic Signalling.

PMID: 25575683 | Indexed in PubMed

Microglia play an important role in synaptic pruning and controlled phagocytosis of neuronal cells during developmental stages. However, the mechanisms that regulate these functions are not completely understood. The present study was designed to investigate the role of purinergic signalling in microglial migration and phagocytic activity during post-natal brain development. One-day-old BALB/c mice received lipopolysaccharide (LPS) and/or a purinergic analogue (2-methylthioladenosine-5'-diphosphate; 2MeSADP), intracerebroventrically (i.c.v.). Combined administration of LPS and 2MeSADP resulted in activation of microglia as evident from increased expression of ionised calcium-binding adapter molecule 1 (Iba1). Activated microglia showed increased expression of purinergic receptors (P2Y2, P2Y6 and P2Y12). LPS either alone or in combination with 2MeSADP induced the expression of Na(+)/Ca(2+) exchanger (NCX-1) and P/Q-type Ca(2+) channels along with MARCKS-related protein (MRP), which is an integral component of cell migration machinery. In addition, LPS and 2MeSADP administration induced the expression of microglial CD11b and DAP12 (DNAX-activation protein 12), which are known to be involved in phagocytosis of neurons during development. Interestingly, administration of thapsigargin (TG), a specific Ca(2+)-ATPase inhibitor of endoplasmic reticulum, prevented the LPS/2MeSADP-induced microglial activation and migration by down-regulating the expression of Iba1 and MRP, respectively. Moreover, TG also reduced the LPS/2MeSADP-induced expression of CD11b/DAP12. Taken together, the findings reveal for the first time that Ca(2+)-mediated purinergic receptors regulate the migration and phagocytic ability of microglia during post-natal brain development.

The roles of microglia in neural remodeling during retinal degeneration.

PMID: 34693982 | Indexed in PubMed

Retina remodeling is a consequence of many retinal degenerative diseases that are characterized by progressive photoreceptor death. Retina remodeling involves a series of complex pathological processes, consisting of photoreceptor degeneration and death, as well as retinal cell reprogramming and "rewiring". This rewiring alters retinal neural circuits that are centered on synaptic connections and lead to widespread death of retinal cells. Retinal remodeling, especially inner retinal remodeling, is the major factor that limits the effectiveness of various treatment strategies, including cell therapy; thus, it is important to elucidate the mechanisms involved in retinal remodeling during retinal degeneration. Microglia are the dominant immune cells in the retina. Microglia monitor the retinal microenvironment, are activated following retinal injury or degeneration, have powerful phagocytosis capabilities, and play a critical role in synaptic pruning during central neural system development. Analogously, microglia have been found to participate in the clearance of synaptic elements in a complement-dependent manner in the classic retinitis pigmentosa (RP) model, Royal College of Surgeons (RCS) rats, and retard the formation of ectopic neuritogenesis and the deterioration of visual function during retinal degeneration. Since previous research on microglia has rarely concentrated on synaptic remodeling during retinal degeneration, summarizing the microglial mechanisms involved in retinal remodeling is necessary in order to design compounds targeting microglia and retinal remodeling that might be promising therapeutic strategies for treating retinal degeneration.

Prenatal infection leads to ASD-like behavior and altered synaptic pruning in the mouse offspring.

PMID: 27697456 | Indexed in PubMed

Environmental challenges to the maternal immune system during pregnancy have been associated with an increase in the frequency of neurodevelopmental disorders such as Autism Spectrum Disorders (ASD) appearing in the offspring. Microglia, the brain's resident immune-cells, are now known to be critically involved in normal brain development, shaping connections between neurons by pruning superfluous synaptic spines. Our aim was to investigate whether maternal infection during critical stages of gestation compromises the role of microglia in sculpting neuronal circuits. Using a mouse model of maternal immune activation (MIA) induced by bacterial Lipopolysaccharide (LPS), we assayed the offspring's behavior during postnatal development. Additionally, we quantified spines within the offspring's brain and assessed alterations in some molecular signals involved in pruning. LPS-induced MIA led to behavioral changes relevant to ASD in the offspring in the absence of gross neurological problems. Prenatal LPS resulted in a significant increase in the number of spines in the granule cells of the dentate gyrus, as well as a reduction in hippocampal expression of the fractalkine microglial receptor (CX3CR1), involved in mediating the pruning process in the offspring. Interestingly, these changes were only noted in the male progeny of the LPS challenged dams. These results provide an early indicator that microglial function is altered in the brain of offspring from immune challenged mothers and that the effects in the brain appear to be specific along sex lines.

Microglia roles in synaptic plasticity and myelination in homeostatic conditions and neurodevelopmental disorders.

PMID: 31058364 | Indexed in PubMed

Microglia are the immune cells of the brain, involved in synapse formation, circuit sculpting, myelination, plasticity, and cognition. Being active players during early development as well as in adulthood, microglia affect other cells directly by their long processes and unique receptors and indirectly by secreting growth factors and cytokines. In this review, we discuss the roles of microglia in neurodevelopmental disorders, synaptic plasticity, myelination, and homeostatic conditions throughout human and mouse development. Within these processes, we specifically focus on the contribution of altered microglial interactions with neurons and oligodendrocytes, altered cytokine and growth factor activities, and alterations in the complement system. We conclude by highlighting future perspectives and providing an overview of future research on microglia.

Maternal immune activation altered microglial immunoreactivity in the brain of postnatal day 2 rat offspring.

PMID: 30256454 | Indexed in PubMed

Microglia, the resident immune cells of the central nervous system, play critical roles in neurodevelopment, synaptic pruning, and neuronal wiring. Early in development, microglia migrate via the tangential and radial migration pathways to their final destinations and mature gradually, a process that includes morphological changes. Recent research has implicated microglial abnormality in the etiology of schizophrenia. Since prenatal exposure to viral or bacterial infections due to maternal immune activation (MIA) leads to increased risk of schizophrenia in the offspring during adulthood, the present study systematically investigated how MIA induced by polyinosinic:polycytidylic acid (a mimic of viral double-stranded RNA) affected microglial immunoreactivity along the migration and maturation trajectories in the brains of male and female rat offspring on postnatal day (PND) 2. The immunohistochemistry revealed significant changes in the density of IBA-1 immunoreactive cells in the corpus callosum, somatosensory cortex, striatum, and the subregions of the hippocampus of the MIA offspring. The male and female MIA offspring displayed markedly altered microglial immunoreactivity in both the tangential and radial migration, as well as maturation, pathways when compared to their sex- and age-matched controls as evidenced by morphology-based cell counting. Given the important roles of microglia in synaptic pruning and neuronal wiring and survival, these changes may lead to structural and functional neurodevelopmental abnormalities, and so contribute to the functional deficits observed in juvenile and adult MIA offspring. Future research is required to systematically determine how MIA affects microglial migration and maturation in rat offspring.

Vaccinia-related kinase 2 plays a critical role in microglia-mediated synapse elimination during neurodevelopment.

PMID: 31050055 | Indexed in PubMed

During postnatal neurodevelopment, excessive synapses must be eliminated by microglia to complete the establishment of neural circuits in the brain. The lack of synaptic regulation by microglia has been implicated in neurodevelopmental disorders such as autism, schizophrenia, and intellectual disability. Here we suggest that vaccinia-related kinase 2 (VRK2), which is expressed in microglia, may stimulate synaptic elimination by microglia. In VRK2-deficient mice (VRK2KO), reduced numbers of presynaptic puncta within microglia were observed. Moreover, the numbers of presynaptic puncta and synapses were

abnormally increased in VRK2KO mice by the second postnatal week. These differences did not persist into adulthood. Even though an increase in the number of synapses was normalized, adult VRK2KO mice showed behavioral defects in social behaviors, contextual fear memory, and spatial memory.

Microglia: Key players in neurodevelopment and neuronal plasticity.

PMID: 29197626 | Indexed in PubMed

Microglia are the primary innate immune cells in the CNS. Since their initial discovery and characterization, decades of research have revealed their unique roles not only in maintaining immune homeostasis, but also being indispensable to brain development and cognitive function. As such, microglia drive synaptogenesis, synaptic pruning, neurogenesis and neuronal activity. Microglia-specific mutations are implicated in several neurodevelopmental disorders, and dysregulation of microglial function is strongly linked to several pathologies, including cognitive decline and Alzheimer's disease. Importantly, developmental insults can lead to long-term changes in microglial function that may compromise the ability of the adult brain to fight infections and process cognitive information. Adult lifestyle or injury can also lastingly influence microglial morphology and function. Here we highlight key research on microglia's role in neuronal plasticity across the lifespan.

[Synaptic plasticity and synaptic reorganization regulated by microglia].

PMID: 25069260 | Indexed in PubMed

Microglia are generally believed to be brain macrophages, which become phagocytic cells after cellular activation in response to inflammation or injury in the brain. However, accumulating evidence suggests that microglia modulate neurotransmission and synaptic plasticity by secretion of several soluble factors. Importantly, microglia secrete glycine to enhance NMDA receptor-mediated responses and hippocampal long-term potentiation, a cellular basis of learning and memory. Although the expression of NMDA receptors was also observed in microglia, NMDA receptor-mediated responses were not induced in microglia. This suggests that NMDA receptors expressed in microglia are not functional. Besides the modulation of synaptic transmission, microglia also play an important role in synaptic remodeling by the pruning of unnecessary synapses and axon terminals during the postnatal developmental stage and adaptation to novel environments even in the healthy brain. Furthermore, we have recently found that clock genes in microglia drive P2Y₁₂R and cathepsin S to regulate diurnal change in the synaptic activity. Therefore, defects in these microglial functions may eventually result in several brain diseases including neuropsychiatric disorders.

Microglia and Sensitive Periods in Brain Development.

PMID: 34463934 | Indexed in PubMed

From embryonic neuronal migration to adolescent circuit refinement, the immune system plays an essential role throughout central nervous system (CNS) development. Immune signaling molecules serve as a common language between the immune system and CNS, allowing them to work together to

modulate brain function both in health and disease. As the resident CNS macrophage, microglia comprise the majority of immune cells in the brain. Much like their peripheral counterparts, microglia survey their environment for pathology, clean up debris, and propagate inflammatory responses when necessary. Beyond this, recent studies have highlighted that microglia perform a number of complex tasks during neural development, from directing neuronal and axonal positioning to pruning synapses, receptors, and even whole cells. In this chapter, we discuss this literature within the framework that immune activation during discrete windows of neural development can profoundly impact brain function long-term, and thus the risk of neurodevelopmental and neuropsychiatric disorders. In this chapter, we review three sensitive developmental periods - embryonic wiring, early postnatal synaptic pruning, and adolescent circuit refinement - in order to highlight the diversity of functions that microglia perform in building a brain. In reviewing this literature, it becomes obvious that timing matters, perhaps more so than the nature of the immune activation itself; largely conserved patterns of microglial response to diverse insults result in different functional impacts depending on the stage of brain maturation at the time of the challenge.
